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	TASK 53-00-11-283-004 - Aerodynamic Requirements for Allowable Damage and Repairs	

TASK 53-00-11-283-004

Aerodynamic Requirements for Allowable Damage and Repairs

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

WARNING: BE CAREFUL WHEN YOU APPLY CHROMIC ACID ANODIZING (CAA) AND CHEMICAL CONVERSION COATING. THE CHEMICALS USED ARE POISONOUS AND CAN CAUSE INJURY. SEE THE MATERIAL SAFETY DATA SHEETS.

CAUTION: MAKE SURE THAT THE EFFECTIVITY OF THE MODIFICATION IS APPLICABLE FOR YOUR AIRCRAFT.

1. Task Supporting Data

A. General

This task contains allowable damage and repair limits around static probe ports, pitot probes, total air temperature probes and angle of attack probes.

NOTE: For definition of allowable damage, refer to [Chapter 51-11-11](#).

NOTE: For damage / repair data recording, refer to [Chapter 51-11-15](#).

NOTE: For repair of minor damage, refer to [Chapter 51-73-00](#).

NOTE: For repair of corroded areas, refer to [Chapter 51-74-00](#).

- (1) These limits are in compliance with the reduced vertical separation minima (RVSM) requirements.
- (2) These limits should be used in conjunction with the allowable damage and repairs for the affected areas given in the relevant task, which address the structural aspect. The lowest value found, using both the aerodynamic and the structural limits is to be considered as the maximum allowable.
The determined value may be higher than the limitations given in [Chapter 51-10-00](#) Aerodynamic Smoothness. In this case, it is mandatory to rework or repair the damage to be within these requirements.
- (3) In case a damage affects a port / probe / sensor and another port / probe / sensor simultaneously, the lowest determined value of the two allowable limits must be used as the maximum allowable limit to ensure good aerodynamic performance.

B. Area of Applicability

This paragraph is intentionally left blank.

C. Weight Variant Applicability

NOTE: This task is applicable for all Weight Variants listed in the SRM Introduction for the related Aircraft.

D. Limits/Restrictions

This task contains aerodynamic aspects only.

In case a damage (e.g. scratch) is within the negligible damage limits given in [Chapter 51-73-11](#) Paragraph 4, this task can be disregarded.

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- E. Repair Category
This task is category A.

NOTE: For definition of the repair categories, refer to [Chapter 51-11-14](#).

- F. Inspections
This paragraph is intentionally left blank.

- G. Job Set-Up Information

- (1) Material Reference

ITEM	NOMENCLATURE	QTY	MATERIAL/REMARKS
-	Surface Pretreatment - Light Alloys Chemical Conversion Clear	AR	Material No. 10ABD1 (Refer to CML)

- (2) References

[Chapter 51-10-00](#)
[Chapter 51-11-00](#)
[Chapter 51-11-11](#)
[Chapter 51-11-14](#)
[Chapter 51-11-15](#)
[Chapter 51-21-11](#)
[Chapter 51-73-00](#)
[Chapter 51-73-11](#)
[Chapter 51-74-00](#)

2.

- A. Static Probe Ports

- (1) Definition of Areas

- Refer to Figure [005](#) and Figure [006](#).

- (2) Definition of characteristic Parameters for Damages and External Repairs

- Refer to Figure [004](#).

- (3) Aerodynamic Requirements

The characteristic parameters (D and d) given in Figure [004](#), Sheet [1](#) must meet the aerodynamic requirements given in Diagram [001](#) and Diagram [002](#), when the damage or repair is within the areas given in Figure [006](#).

NOTE: Make sure that the internal surfaces of the probe port holes are clean and free of burrs and the outer edge of the probe port holes are clean and free of burrs, raised edges or nicks.

- (a) Requirements for Allowable Damages and External Repairs
The limitation data is given in Diagram [001](#) and Diagram [002](#).

- (b) Additional Requirements

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1 Applicable to the static probe ports from FR23 thru FR24

a Fastener flushness, Skin waviness and Surface roughness:

The area defined by the radius R1 given in Figure [006](#):

- Fastener flushness and Surface roughness (Rt) must not exceed ± 0.08 mm (0.0031 in).
- Skin waviness must not exceed ± 0.4 mm (0.016 in) at a wavelength of 150 mm (5.91 in) centered on the static probe port, on horizontal axis compared to the fuselage theoretical profile.

The crown area defined by the radius R2-R1 given in Figure [006](#):

- Fastener flushness and Surface roughness (Rt) must not exceed ± 0.25 mm (0.0098 in).
- Skin waviness must not exceed ± 0.8 mm (0.031 in) at a wavelength of 300 mm (11.81 in) centered on the static probe port, on horizontal axis compared to the fuselage theoretical profile.

b Damage caused by Lightning Strikes, Nicks and Scratches

1- A total of 12 damages are permitted within the area defined by the radius R2 (given in Figure [006](#)), provided Table [1](#) requirements are fulfilled:

- with a maximum of 2 (of the 12 being permitted) within the area defined by a radius of 15 mm (0.59 in) centered on the static probe port
- with a maximum of 7 (of the 12 being permitted) within the crown area defined by the radius R1 (given in Figure [006](#)) - 15 mm (0.59 in) centered on the static port
- with a maximum of 12 (of the 12 being permitted) within the crown area defined by the radius R2-R1 (given in Figure [006](#)).

AREA	MAXIMUM DIAMETER OR LENGTH OF EACH DAMAGE	MAXIMUM DEPTH OF EACH DAMAGE	MAXIMUM THICKNESS OF THE PROTRUDING PARTS (COMING FROM MELTED METAL)
The area defined by a radius of 15 mm (0.59 in) centered on the static probe port	4 mm (0.16 in)	0.15 mm (0.0059 in)	0.05 mm (0.002 in)
The crown area defined by the radius R1 (given in Figure 006) - 15 mm (0.59 in) centered on the static port	6 mm (0.24 in)	0.2 mm (0.008 in)	0.1 mm (0.004 in)

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The crown area defined by the radius R2-R1 given in Figure 006	10 mm (0.39 in)	0.4 mm (0.016 in)	0.2 mm (0.008 in)
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Table 1 - Damage Dimensions

2- In addition to the previous paragraph, it is permitted to have un-reworked scratches as follows:

- a) A total of 10 un-reworked scratches are permitted within the area defined by the radius R1 (given in Figure [006](#)):
- With a maximum of 5 (of the 10 being permitted) within the area defined by a radius of 15 mm (0.59 in) centered on the static probe port, with depth D # 0.02 mm (0.0008 in), length # 20 mm (0.79 in) and cumulative length # 40 mm (1.57 in)
 - With a maximum of 10 (of the 10 being permitted) within the crown area defined by the radius R1 (given in Figure [006](#)) - 15 mm (0.59 in) centered on the static port, with depth D # 0.04 mm (0.0016 in), length # 30 mm (1.18 in) and cumulative length # 80 mm (3.15 in).
- b) In addition, a total of 3 un-reworked scratches are permitted within the crown area defined by the radius R1 (given in Figure [006](#)) - 15 mm (0.59 in) centered on the static port:
- With a maximum of 3 (of the 3 being permitted) with depth D # 0.02 mm (0.0008 in) whatever its length
 - With a maximum of 3 (of the 3 being permitted) with depth D # 0.1 mm (0.004 in) and length # 50 mm (1.97 in).
- c) All the burrs or protruding parts must be removed. Apply surface pretreatment (Material No. [10ABD1](#)) for scratches in the aluminum alloys, refer to [Chapter 51-21-11](#).

2 Applicable to the standby static probe port blanking plate from FR3 thru FR4

- a Mismatches (Steps), Skin waviness and Surface roughness:
The acceptable mismatch between the head of the 9 pins and the blanking plate must not exceed ± 0.15 mm (0.0059 in).
The Surface roughness (Rt) must not exceed ± 0.15 mm (0.0059 in).
The acceptable mismatch between the blanking plate and the skin panel is + 0.55 mm (0.0217 in) with tolerances of:
- ± 0.35 mm (0.0138 in) in the line of flight (IN LOF)
 - ± 0.25 mm (0.0098 in) across line of flight (A LOF).
- Skin waviness must not exceed ± 0.10 mm (0.0039 in) on the blanking plate, on horizontal axis compared to the fuselage theoretical profile.
- b Damage caused by lightning strikes, nicks or scratches:
- 1- A total of 7 damages are permitted on standby static probe port blanking plate, with a maximum of 2 (of the 7 being permitted) within the area defined by a radius of 15 mm (0.59 in) centered on the static probe port, provided:
- The diameter or length of each damage must not exceed 4 mm (0.16 in)
 -

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The maximum depth of each damage must not exceed 0.15 mm (0.0059 in)

- The maximum thickness of the protruding parts (coming from melted metal) of each damage must not exceed 0.05 mm (0.002 in)
- Outer edge of the hole must be free of burrs, raised edges or nicks.

2- In addition to the previous paragraph, it is permitted to have un-reworked scratches as follows:

a) A total of 10 un-reworked scratches are permitted on the standby static probe port blanking plate:

- With a maximum of 5 (of the 10 being permitted) within the area defined by a radius of 15 mm (0.59 in) centered on the static probe port, with depth D # 0.02 mm (0.0008 in), length # 20 mm (0.79 in) and cumulative length # 40 mm (1.57 in)
- With a maximum of 10 (of the 10 being permitted) within the area defined by the standby static probe port blanking plate - a radius of 15 mm (0.59 in) centered on the static port, with depth D # 0.04 mm (0.0016 in), length # 30 mm (1.18 in) and cumulative length # 80 mm (3.15 in).

b) In addition, a total of 3 un-reworked scratches are permitted within the area defined on the standby static probe port blanking plate - a radius of 15 mm (0.59 in) centered on the static port:

- With a maximum of 3 (of the 3 being permitted) with depth D # 0.02 mm (0.0008 in) whatever its length
- With a maximum of 3 (of the 3 being permitted) with depth D # 0.1 mm (0.004 in) and length # 50 mm (1.97 in).

c) All the burrs or protruding parts must be removed. Apply surface pretreatment (Material No. [10ABD1](#)) for scratches in the aluminum alloys, refer to [Chapter 51-21-11](#).

(c) Repairs

If the damage is more than the requirements and limitations given in Step [2.A.\(3\).\(a\)](#), repair the damage according to the aerodynamic requirements given in Diagram [001](#) and Diagram [002](#) and the requirements given below:

- 1 For area defined by radius R1 given in Figure [006](#):
 - No external repair doubler is permitted. Only a flush doubler must be installed in this area.
- 2 For area 'A' given in Figure [005](#):
 - Leading edge and trailing edge of an external repair doubler are not permitted.
- 3 For area 'B' given in Figure [005](#):
 - The chamfer shown in Figure [003](#) is mandatory on the external repair doubler.
 - Only one external repair doubler and/or damage is permitted in this area.

NOTE: If the damage cannot be repaired within the above requirements, contact AIRBUS.

B. Angle of Attack Probes

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- (1) Definition of Areas
 - Refer to Figure [001](#).
- (2) Definition of Characteristic Parameters for Damages and External Repairs
 - Refer to Figure [004](#).
- (3) Aerodynamic Requirements

NOTE: The characteristic parameters (D and d) given in Figure [004](#), Sheet [1](#) must meet the aerodynamic requirements given in Diagram [003](#), when the damage or repair is within the areas given in Figure [001](#).

NOTE: After Modification 153213 and/or 153214, the Angle of Attack Conic Cover Plates are non Customized and shall remain grey BAC 707/M9001.

- (a) Requirements for allowable damages and external repairs
 - The limitation data is given in Diagram [003](#).
- (b) Additional requirements
 - 1 For areas defined by Figure [001](#), radius R1:
 - No repair doubler edge is permitted in this area.
 - 2 For hatched area defined by Figure [001](#), radius R2 - radius R1:
 - If more than one repair and/or damage in this area, contact AIRBUS.
 - The chamfer defined in Figure [003](#) is mandatory on the repair doubler.

C. Pitot Probes and Total Air Temperature Probes

- (1) Definition of Areas
 - Refer to Figure [002](#).
- (2) Definition of Characteristic Parameters for Damages and External Repairs
 - Refer to Figure [004](#).
- (3) Aerodynamic Requirements

NOTE: The characteristic parameters (D and d) given in Figure [004](#), Sheet [1](#) must meet the aerodynamic requirements given in Diagram [004](#), when the damage or repair is within the areas given in Figure [002](#).

- (a) Requirements for allowable damages and external repairs
 - The limitation data is given in Diagram [004](#).
- (b) Additional requirements
 - 1 For hatched area defined by Figure [002](#), Radius R1 and Radius R2:
 - If more than one repair and/or damage in this area, contact AIRBUS.
 - The chamfer defined in Figure [003](#) is mandatory on the repair doubler.

D. Measurement of Skin Waviness

- (1) For measurement of skin waviness, refer to [Chapter 51-11-00](#).
 -  [Diagram 001 / 53-00-11-283-004 \(sheet 1\) Static Probe Ports - Limitation Data](#)
 -  [Diagram 002 / 53-00-11-283-004 \(sheet 1\) Static Probe Ports - Limitation Data](#)

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-  [Figure 001 / 53-00-11-283-004 \(sheet 1\) Angle of Attack Probes - Definition of Areas](#)
-  [Diagram 003 / 53-00-11-283-004 \(sheet 1\) Angle of Attack Probes - Limitation Data](#)
-  [Figure 002 / 53-00-11-283-004 \(sheet 1\) Pitot Probes and Total Air Temperature Probes - Definition of Areas](#)
-  [Diagram 004 / 53-00-11-283-004 \(sheet 1\) Pitot Probes and Total Air Temperature Probes - Limitation Data](#)
-  [Figure 003 / 53-00-11-283-004 \(sheet 1\) Chamfer Data](#)
-  [Figure 004 / 53-00-11-283-004 \(sheet 1\) Definition of Characteristic Parameters](#)
-  [Figure 004 / 53-00-11-283-004 \(sheet 2\) Definition of Characteristic Parameters](#)
-  [Figure 005 / 53-00-11-283-004 \(sheet 1\) Static Probe Ports - Definition of Areas](#)
-  [Figure 006 / 53-00-11-283-004 \(sheet 1\) Static Probe Ports - Definition of Areas](#)

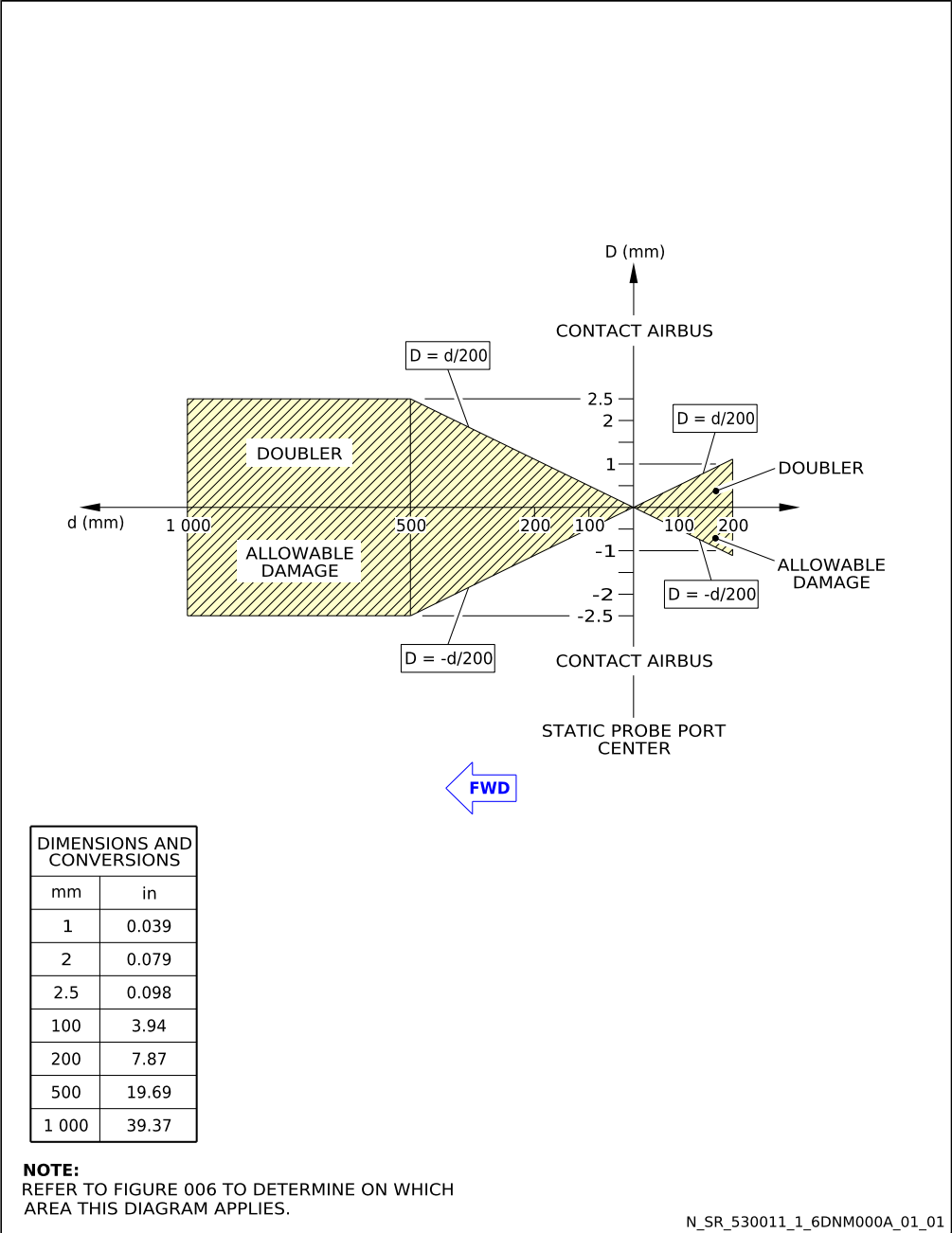
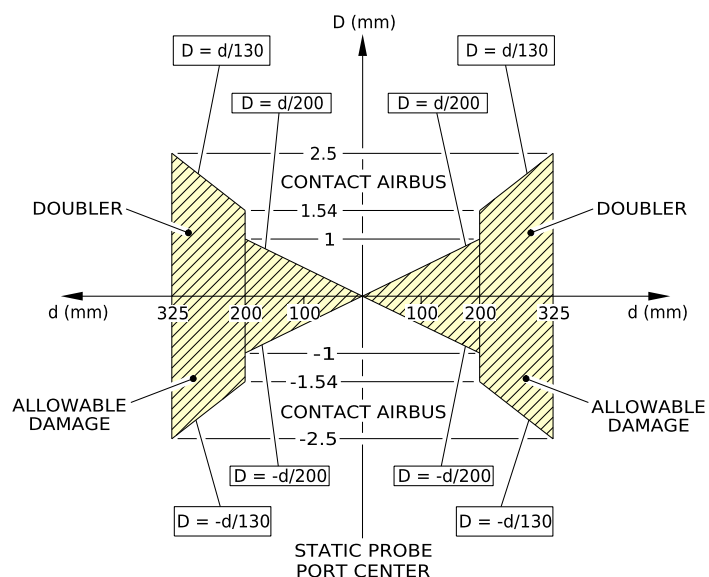


Diagram 001 / 53-00-11-283-004 (SHEET 1) - Static Probe Ports - Limitation Data

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DIMENSIONS AND CONVERSIONS	
mm	in
1	0.039
1.54	0.061
2.5	0.098
100	3.94
200	7.87
325	12.795

NOTE:
REFER TO FIGURE 006 TO DETERMINE ON WHICH
AREA THIS DIAGRAM APPLIES.

N SR 530011 1 6DOM000A 01 01

Diagram 002 / 53-00-11-283-004 (SHEET 1) - Static Probe Ports - Limitation Data

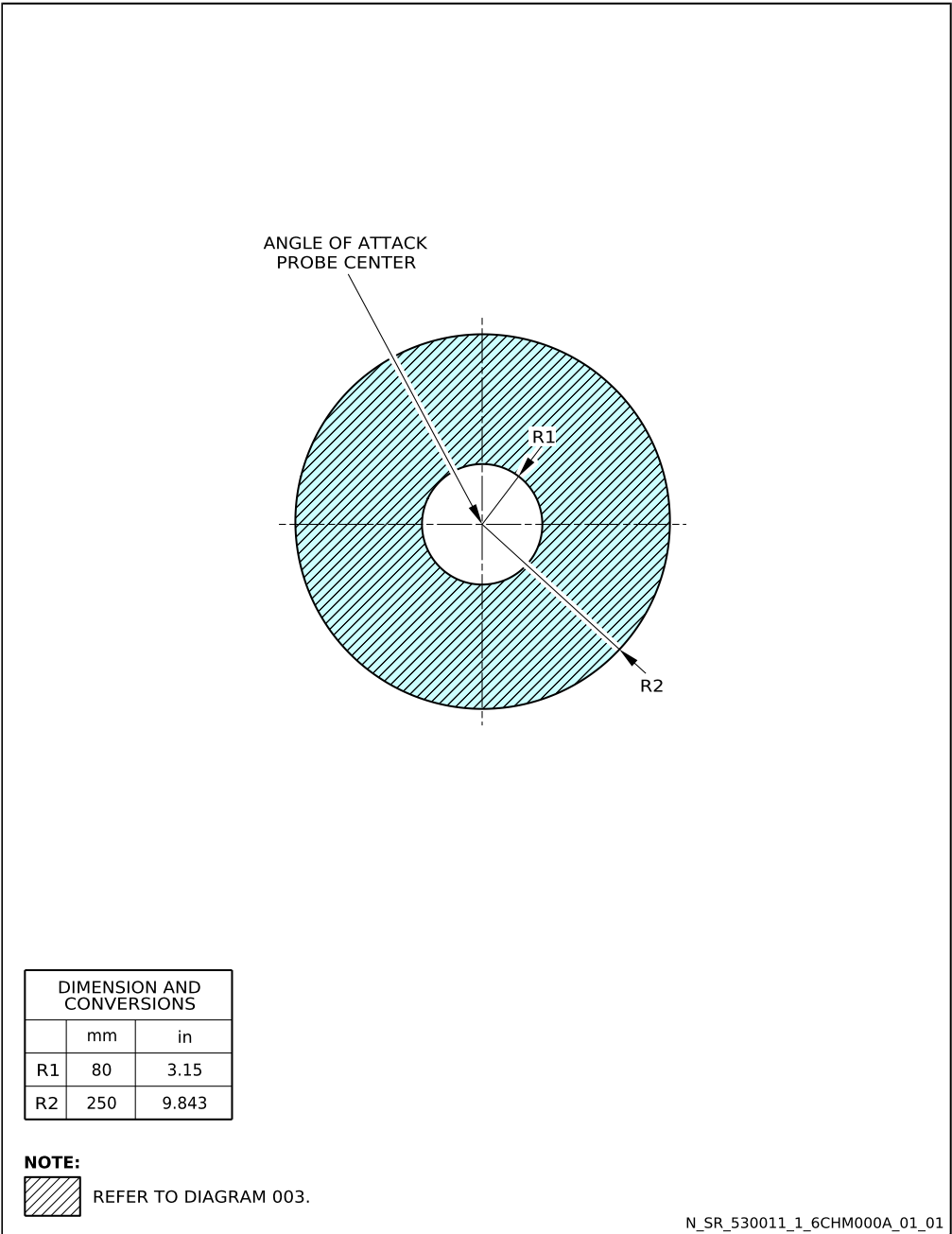


Figure 001 / 53-00-11-283-004 (SHEET 1) - Angle of Attack Probes - Definition of Areas

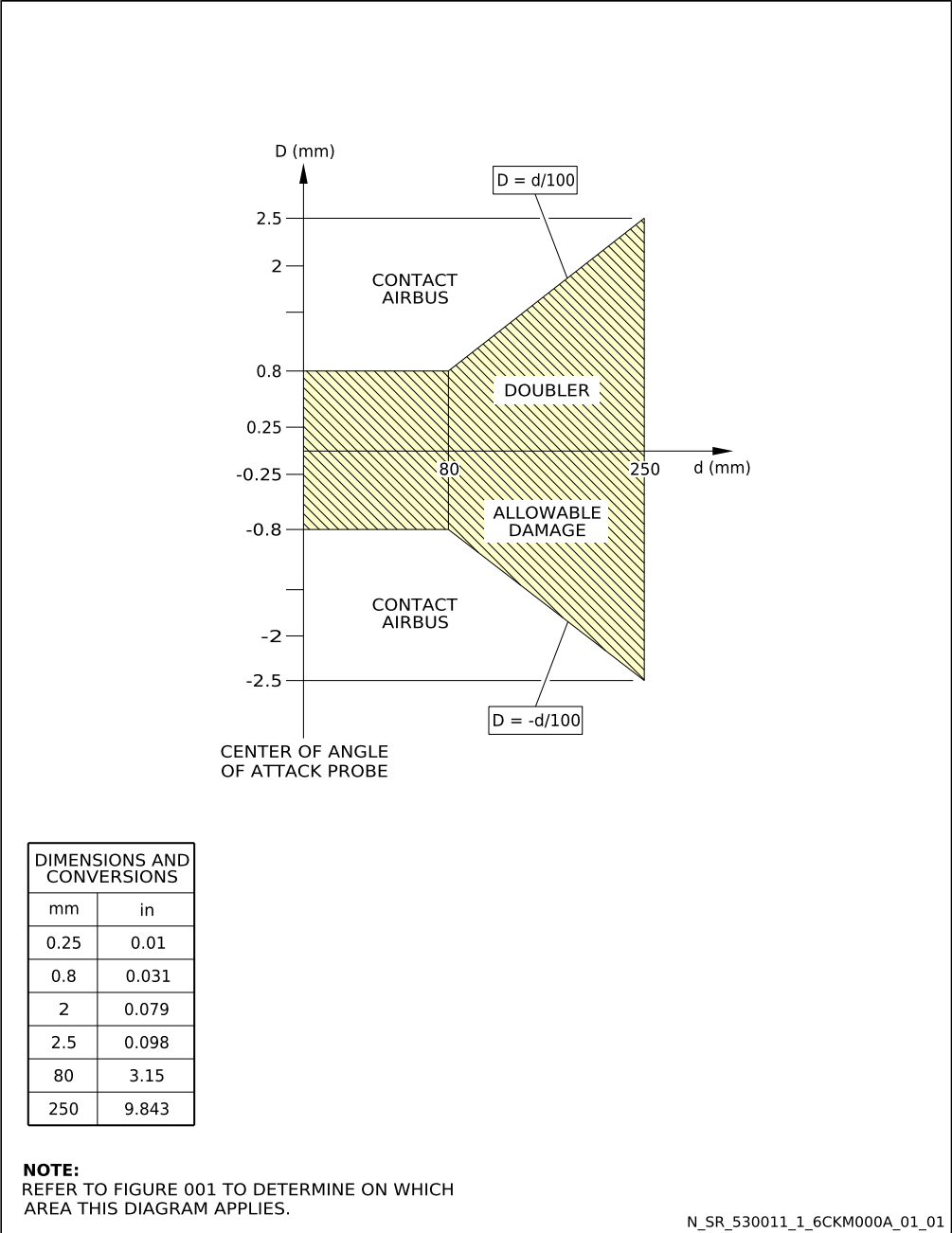


Diagram 003 / 53-00-11-283-004 (SHEET 1) - Angle of Attack Probes - Limitation Data

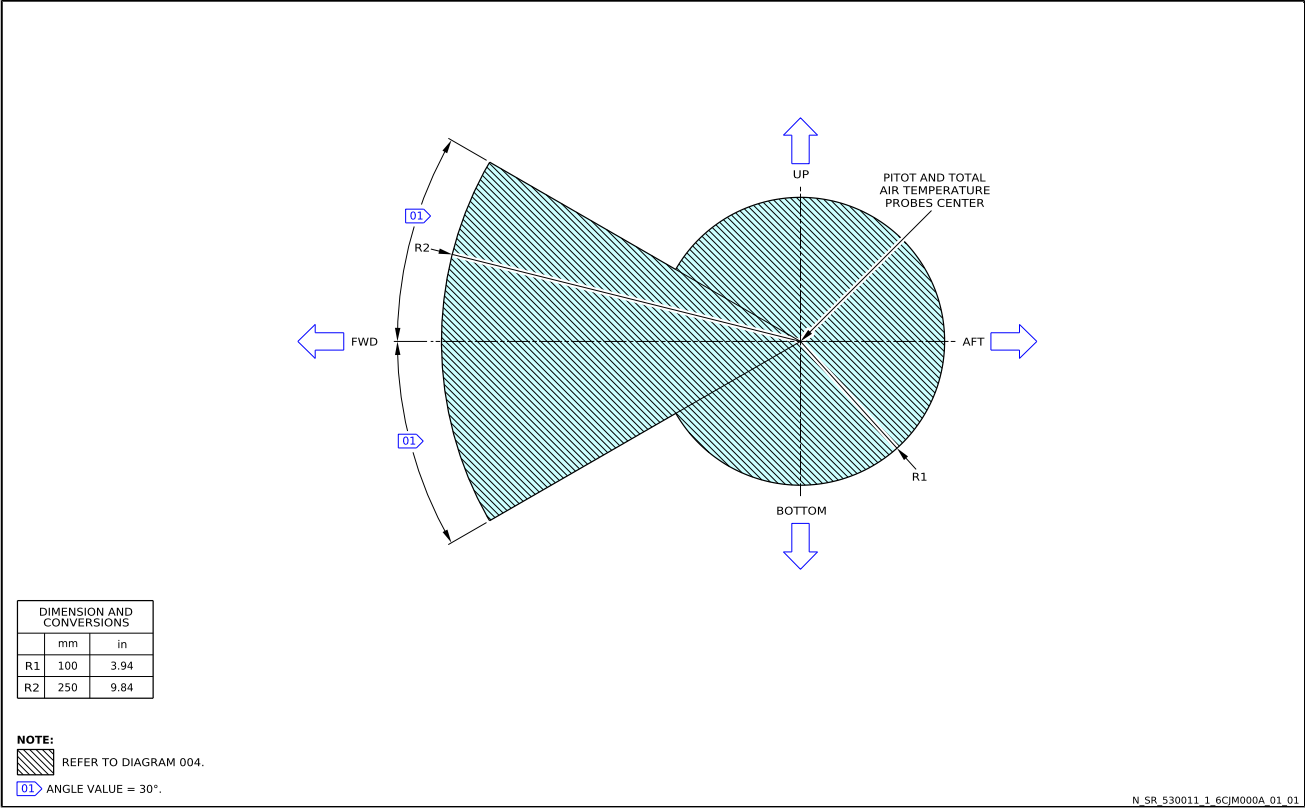


Figure 002 / 53-00-11-283-004 (SHEET 1) - Pitot Probes and Total Air Temperature Probes - Definition of Areas

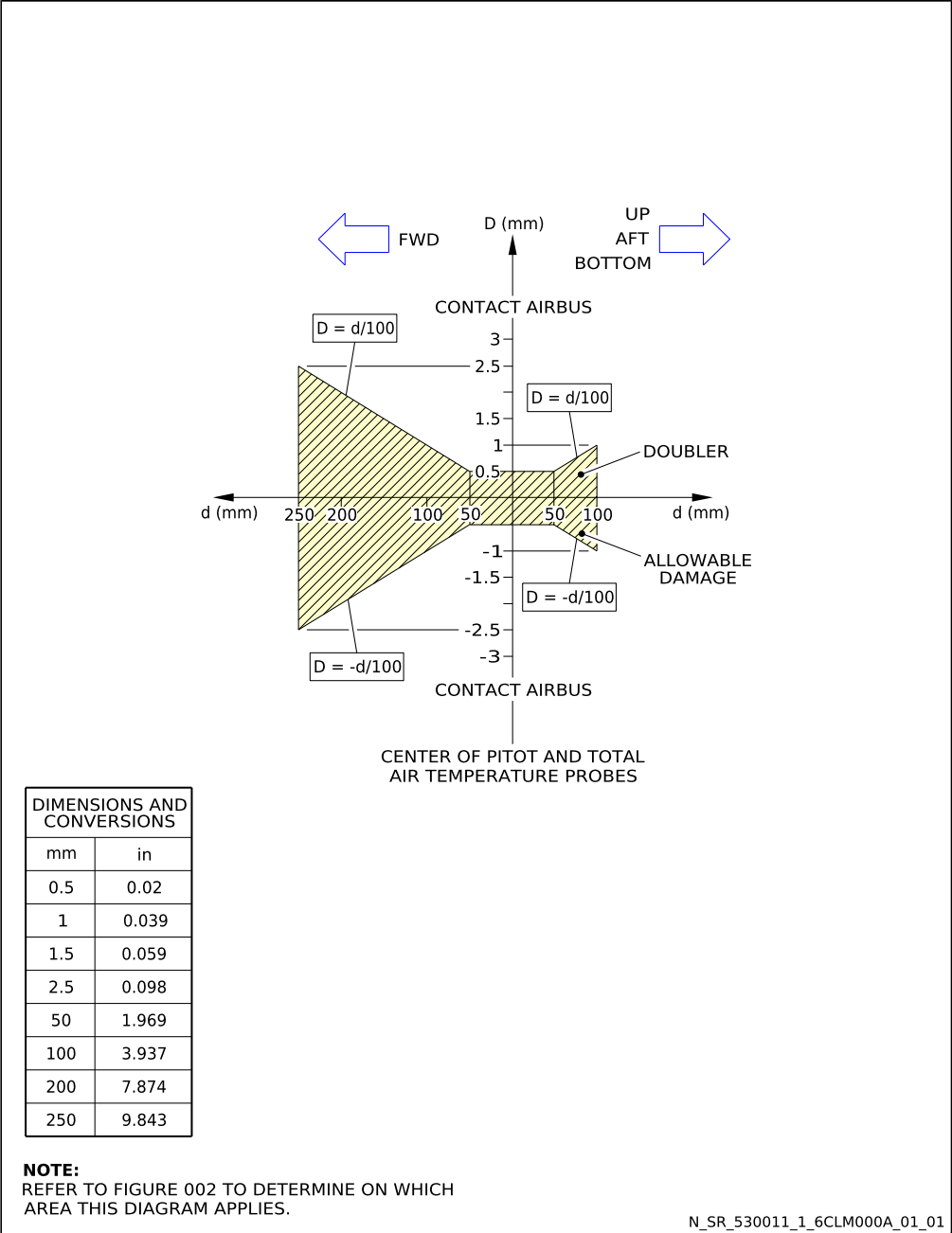


Diagram 004 / 53-00-11-283-004 (SHEET 1) - Pitot Probes and Total Air Temperature Probes - Limitation Data

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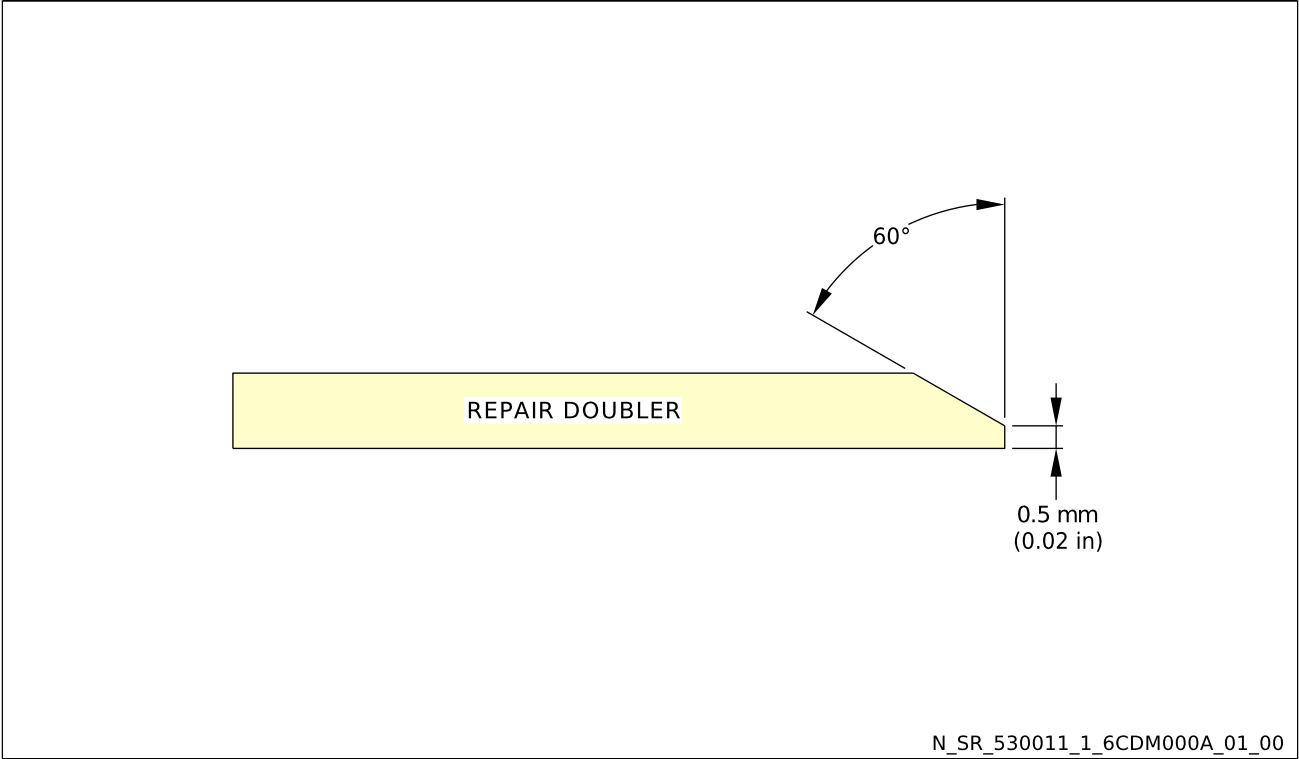


Figure 003 / 53-00-11-283-004 (SHEET 1) - Chamfer Data

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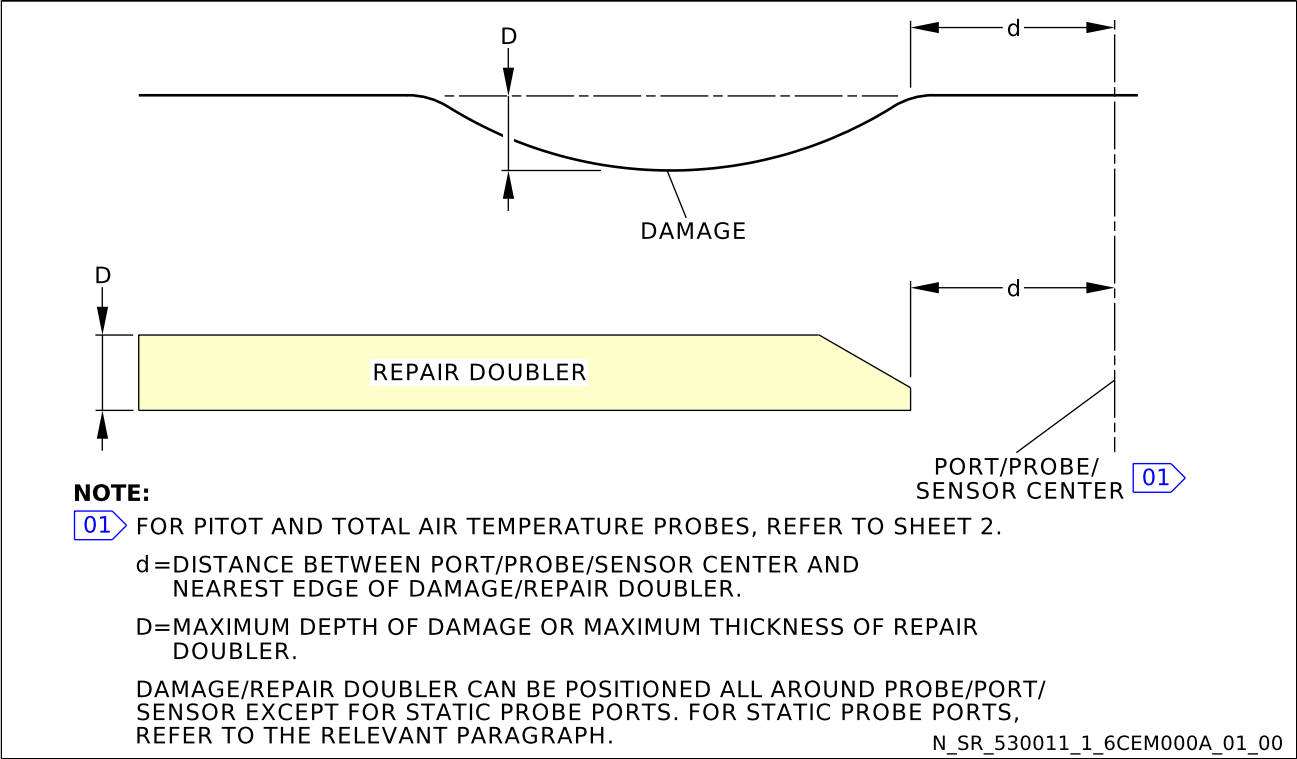


Figure 004 / 53-00-11-283-004 (SHEET 1/2) - Definition of Characteristic Parameters

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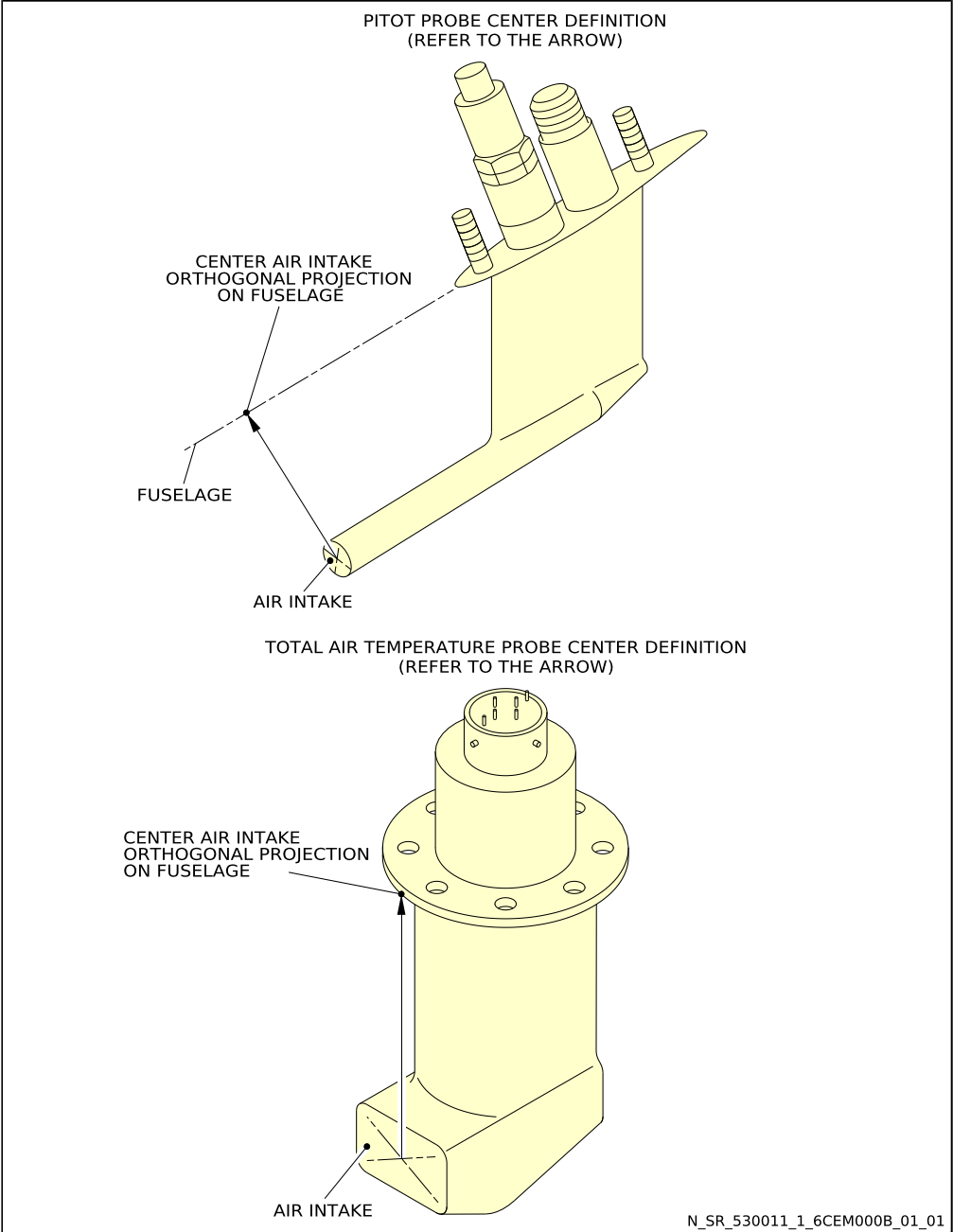


Figure 004 / 53-00-11-283-004 (SHEET 2/2) - Definition of Characteristic Parameters

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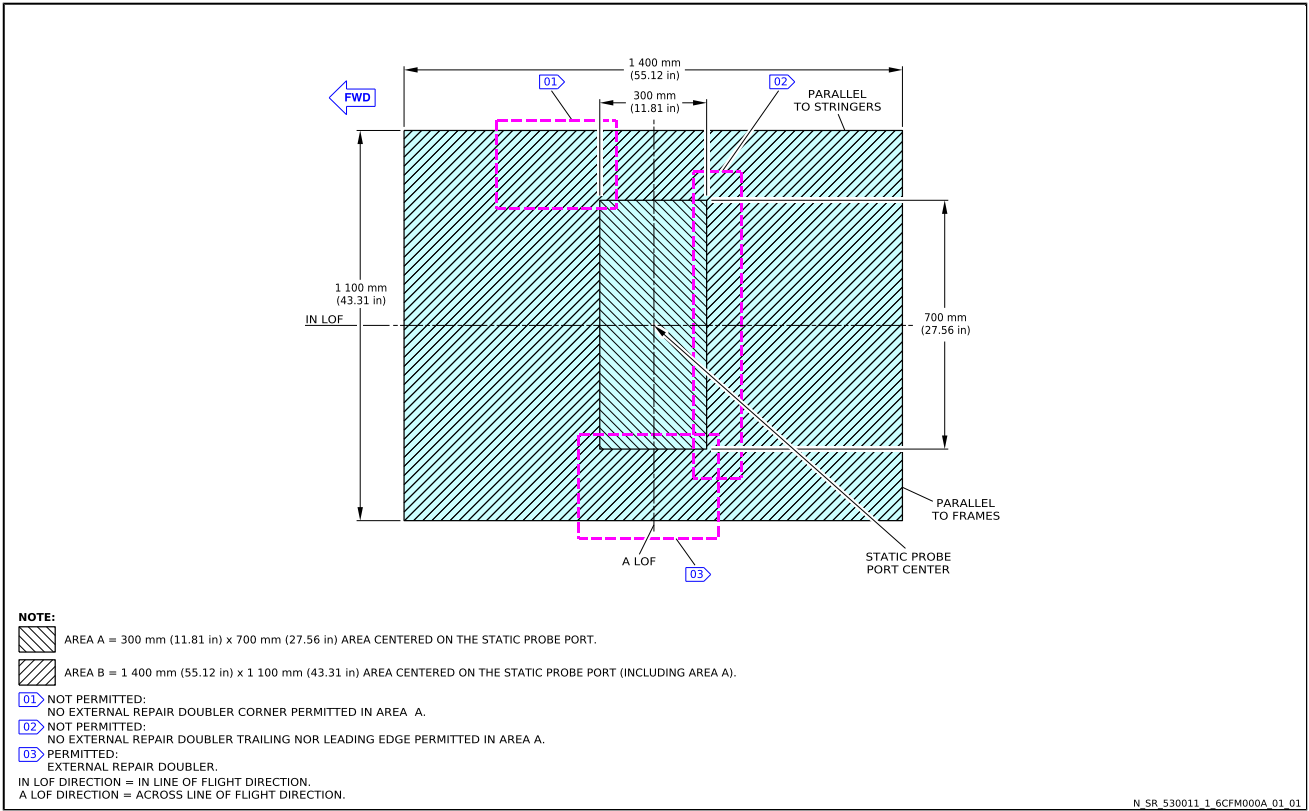


Figure 005 / 53-00-11-283-004 (SHEET 1) - Static Probe Ports - Definition of Areas

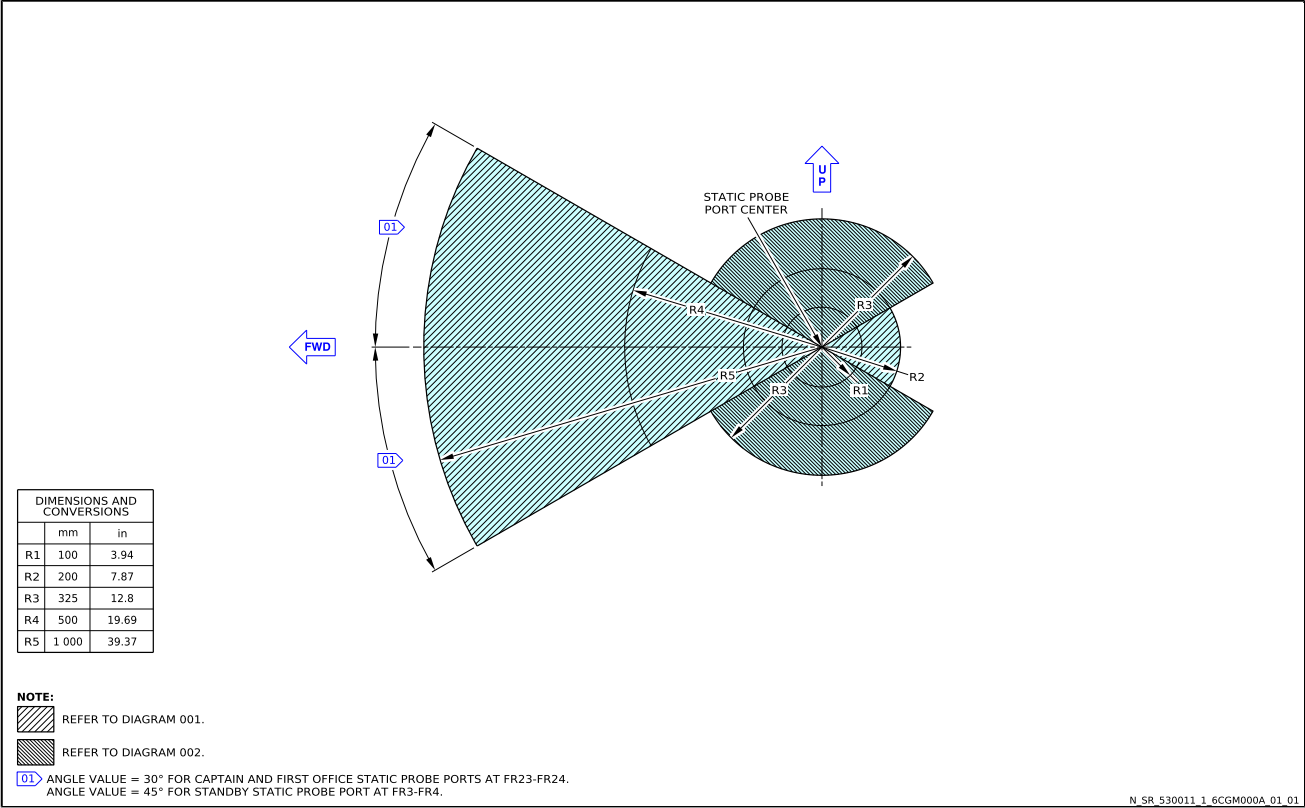


Figure 006 / 53-00-11-283-004 (SHEET 1) - Static Probe Ports - Definition of Areas